Hypertension

ORIGINAL ARTICLE

Cardiometabolic Burden in Bilateral Macronodular Adrenal Disease With Primary Aldosteronism

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BACKGROUND: Bilateral macronodular adrenocortical disease is often linked to autonomous cortisol secretion, but may also present with primary aldosteronism (PA).

METHODS: This international (Europe, United States, Asia) retrospective cohort study included adults with radiological evidence of bilateral macronodular adrenocortical disease and biochemically confirmed PA. The primary endpoints was major adverse cardiovascular events (MACE); secondary end points included cardiometabolic comorbidities and surgical outcomes per PA surgical outcome criteria.

RESULTS: Two hundred forty-nine patients from 41 centers in 12 countries were included (median age, 55 years; 62% male). Median hypertension duration at PA diagnosis was 9.9 years. Among 178 tested, 52% had cortisol cosecretion and 47% isolated PA. At baseline, 56% had metabolic comorbidities, and 16% had ≥1 MACE. Patients with MACE were older, more often male, had longer hypertension duration, and higher diabetes rates. Eighty-nine patients underwent adrenalectomy: 50 without MRA (mineralocorticoid receptor antagonists), 38 with MRA, and 1 with steroidogenesis inhibitors. One hundred twenty-four patients received continued MRA without adrenalectomy or steroidogenesis inhibitors. Adrenal venous sampling showed lateralized PA in 89% of surgical versus 19% of MRA-treated patients ($P<0.001$). Over a median follow-up of 36 (MRA) and 18 months (surgery; $P=0.2$), MACE occurred in 8% and 6%, respectively ($P=1.0$). Blood pressure and organ damage were similar, but more MRA-treated patients needed ≥3 antihypertensives (MRA: 48% versus adrenalectomy: 14%; $P<0.001$). Among operated patients, complete clinical and biochemical success was 26% and 71%, respectively.

CONCLUSIONS: Bilateral macronodular adrenocortical disease with PA carries a high cardiometabolic burden. Early detection and precise subtyping are key to optimizing management and preventing target organ damage. ***(Hypertension.* 2026;83:1364–1375. DOI: 10.1161/HYPERTENSIONAHA.125.25507.)** • **Supplement Material**.

Key Words: adrenalectomy ■ blood pressure ■ hyperplasia ■ mineralocorticoid receptor antagonists ■ prevalence ■ renin

Bilateral macronodular adrenocortical disease (BMAD), formerly referred to as primary bilateral macronodular adrenal hyperplasia, is a rare cause of Cushing syndrome and, more frequently, of mild autonomous cortisol secretion.[1–6] With increasing utilization of cross-sectional imaging, BMAD detection has risen, though the true prevalence remains unclear due to the absence of well-defined diagnostic criteria. BMAD is usually diagnosed by at least 1 macronodule (≥10 mm) on both adrenal glands, regardless of hormonal dysfunction. The size criterion distinguishes BMAD from bilateral micronodular adrenocortical disease (with nodules <10 mm).[7,8] Available evidence on the prevalence of adrenal incidentalomas derives mainly from autopsy and imaging

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NOVELTY AND RELEVANCE

What Is New?

Cardiometabolic outcomes are reported in patients with bilateral macronodular adrenocortical disease and primary aldosteronism following adrenalectomy or mineralocorticoid receptor antagonist therapy.

What Is Relevant?

Patients with bilateral macronodular adrenocortical disease and primary aldosteronism show high rates of cardiometabolic comorbidities, major cardiovascular events, and prolonged hypertension before primary aldosteronism diagnosis.

Surgical and mineralocorticoid receptor antagonist therapy show similar blood pressure and organ damage outcomes, though mineralocorticoid receptor antagonist–treated patients require more antihypertensives. Adrenal venous sampling is essential for optimal treatment selection.

Clinical/pathophysiological implications?

Early diagnosis through broader primary aldosteronism screening, precise subtyping via adrenal venous sample, and targeted treatment are essential to improve outcomes.

Nonstandard Abbreviations and Acronyms

AVS	adrenal venous sampling
BMAD	bilateral macronodular adrenocortical disease
DDD	defined daily dose
DST	1 mg dexamethasone suppression test
iPA	isolated primary aldosteronism
MACE	major cardiovascular event
MRA	mineralocorticoid receptor antagonists
PA	primary aldosteronism
PASO	primary aldosteronism surgery outcome

studies. Autopsy data indicate that unsuspected adrenal masses are present in about 2% of cases (range between 1% and 8.7%). Imaging studies describe a prevalence of roughly 3% around the age of 50, which can rise to nearly 10% in older individuals.[3,9,10] The frequency increases progressively with age, and at the time of diagnosis, bilateral involvement is observed in ≈15% to 20% of patients.[11–13]

Among these, BMAD has been reported in up to one-third of patients with adrenal incidentalomas associated with mild or subclinical hypercortisolism, though its actual prevalence may be underestimated due in part to the fact that some lesions may be nonfunctioning at the time of detection or may secrete steroids that are not routinely screened for, such as androgens or estrogens.[14] Asymmetrical or asynchronous involvement of the 2 adrenal glands further complicates the diagnosis.[15] Bilateral adrenalectomy was historically the preferred treatment for BMAD with relevant cortisol excess, but is now usually reserved for overt Cushing syndrome with marked bilateral adrenal enlargement.[10] Unilateral adrenalectomy is often recommended in cases of adrenal asymmetry with overt Cushing syndrome or mild autonomous cortisol secretion.[4]

Primary aldosteronism (PA), a common form of secondary hypertension (5% to 15% of individuals with arterial hypertension seen in primary care and up to 30% in referral centers),[16–19] is underdiagnosed due to nonspecific features and limited awareness among clinicians. Approximately one-third of PA cases involve lateralized disease, usually due to a dominant aldosterone-producing adenoma, while two-thirds involve nonlateralized disease, typically resulting from bilateral diffuse or nodular adrenal hyperplasia. Both mineralocorticoid and glucocorticoid excess can contribute to increased cardiovascular damage and mortality.[20–25]

To date, only a few case reports and small case series have reported PA in BMAD, probably due to underscreening.[26–29] A recent systematic literature review of these studies identified that 2% to 43% of BMAD cases were associated with PA, more often so in males, and with high rates of metabolic and cardiovascular comorbidities, highlighting the need for increased awareness and targeted treatment.[30] To the best of our knowledge, no previous studies have assessed the cardiovascular risk or optimal treatment approach in patients with BMAD and PA. This study, therefore, aimed to characterize their clinical, hormonal, and radiological features, evaluate cardiovascular risk, and compare biochemical and clinical success rates across different treatment approaches.

METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author on reasonable request.

Study Design and Setting

This international multicenter cohort study was approved by the European Network for the Study of Adrenal Tumours (www.ensat.org) in December 2023. Forty-one centers across 12 countries participated. Standardized, pseudonymized phenotype recording for this study was approved by the local ethics committees of the participating centers. All included patients gave written informed consent, unless waived by local committees. Retrospective data were collected from patient records and, when the data were incomplete or unclear, supplemented through structured telephone interviews. Data were entered into a standardized REDCap spreadsheet between January and April 2024 (4 months). Additional data regarding adrenal venous sampling (AVS) and pharmacological treatment were collected between November and December 2025. These data were used to perform additional predefined analyses requested during the revision process.

Inclusion Criteria and Definitions

Patients were eligible if they had BMAD and PA. BMAD was defined as at least one macronodule (≥10 mm) on both adrenal glands. PA was diagnosed based on the United States[31] or Japanese Endocrine Society guidelines.[32] After discontinuing interfering drugs, screening and confirmatory tests were interpreted per local criteria of each participating center. If complete withdrawal of interfering antihypertensive drugs was not feasible, tests under α1-antagonists, calcium channel blockers, and moxonidine were accepted. The interpretation criteria for screening and confirmatory tests remained consistent between the initial evaluation and follow-up within each center. Plasma renin concentration, plasma renin activity, and aldosterone were measured using validated standard assays routinely employed at each center, in accordance with local laboratory protocols and guidelines recommendations. AVS was used for subtyping PA as per Endocrine Society and European Society of Hypertension guidelines.[31,33] AVS data were collected when available, including selectivity index, lateralization index, and contralateral ratio, both under basal conditions and following ACTH stimulation, according to center-specific protocols. Given the heterogeneity of AVS procedures and interpretation criteria across participating centers, detailed AVS parameters and subgroup comparisons are reported in the Tables S8 through S10. Lateralization was defined using center-specific lateralization index cutoffs, with thresholds ≥2 adopted by some centers and ≥4 by others, reflecting local practice in line with guideline recommendations.[34] For cases falling within the grey zone, a contralateral suppression index <1 was used as an adjunctive criterion to support unilateral disease and surgical eligibility.

Mild autonomous cortisol secretion was diagnosed in patients without overt signs and symptoms of Cushing syndrome who had a morning serum cortisol of >1.8 µg/dL (>50 nmol/L) in the overnight 1 mg dexamethasone suppression test (DST).[35] Cushing syndrome was diagnosed in patients with typical clinical signs and biochemically confirmed cortisol excess following current guidelines recommendations.[35] We excluded patients who did not meet the inclusion criteria or lacked baseline aldosterone, renin, or imaging data. Full criteria are detailed in the study protocol (Supplemental Data).

Variables

Demographic (sex, age, ethnicity, hypertension onset), clinical (body mass index, comorbidities, cardiovascular events), biochemical (screening/confirmatory tests, subtyping), radiological (largest nodule size/side), treatment data, and, when available, gene variant and histological findings were collected. Comorbidities were diagnosed following relevant guidelines at diagnosis and latest follow-up. Blood pressure was obtained from standardized outpatient measurements in a seated position during usual antihypertensive medication use. Hypertension was defined as systolic blood pressure ≥140 mm Hg or diastolic blood pressure ≥90 mm Hg.[36] Antihypertensive drug use was expressed as defined daily doses (DDDs), corresponding to the assumed average maintenance dose per day for each drug when used for its primary indication in adults. DDDs were defined in accordance with the World Health Organization Anatomic Therapeutic Chemical/DDD Index (2017). Calculation of DDDs was performed using a validated online DDD calculation tool (available at: https://github.com/ABurrello/PASO-Predictor/raw/master/00 - PASO Predictor.xlsm

Data were collected at baseline and last follow-up. A complete list of collected data is available in the study protocol (Supplemental Data).

Outcomes

The primary end point was the prevalence of major adverse cardiovascular events (MACE), including coronary artery disease (myocardial infarction, stable or unstable angina pectoris requiring cardiac revascularization, or bypass surgery), major arrhythmias (atrial fibrillation, ventricular tachycardia/fibrillation), heart failure requiring hospitalization, stroke, and cardiovascular-related death. If only the event year were known, the date would be set to June 30. Secondary end points were the prevalence of cardiometabolic comorbidities and clinical/biochemical success after adrenalectomy. Time to MACE was defined as the time from intervention (adrenalectomy or MRA [mineralocorticoid receptor antagonists] therapy) to the first event. MACE were ascertained through a systematic review of medical records by the contributing authors at each participating center, without formal blinded event adjudication. Postoperative (ie, after adrenalectomy) and medical (ie, under MRA therapy) clinical and biochemical success were assessed ≥6 months after adrenalectomy, in agreement with the PASO consensus.[37] Definitions and outcomes are detailed in Table S1.

Statistical Analysis

Continuous variables were reported as mean±SD or median (interquartile range [IQR]), and categorical variables as percentages with available numbers. The Fisher exact test was used for binary variables. Wilcoxon signed-rank test and Mann-Whitney *U* test were used for continuous variables in paired and unpaired samples, respectively. Correlations were analyzed using Spearman correlation. Kaplan-Meier analysis was used to construct survival curves, and Cox regression analysis was used to estimate mortality risk. Subgroup analyses stratified patients by biochemical profile (isolated primary aldosteronism [iPA] versus cortisol cosecretion) and treatment (surgery versus MRA). Multivariable Cox regression analysis was used to identify relevant prognostic variables for the occurrence of new MACE, considering the treatment modality as a time-dependent variable. $P<0.05$ was considered statistically significant. Statistical analyses were conducted using GraphPad Prism version 8.4.2, SPSS version 25, and *R* Version 4.4.2 with the packages survival and survminer.

RESULTS

Clinical Presentation of BMAD With PA at Baseline

A total of 280 patients from 41 centers in 12 countries were recruited, with 249 meeting the inclusion criteria (flow chart of patient selection in Figure S1, center-specific patients and ethnicity in Table S2). Most patients were male (154/249, 62%) and had a median age of 55.0 years (50.0–62.5; Table 1). Based on a pathological 1 mg-DST of >1.8 µg/dL (>50 nmol/L), 93 patients (37%) had cortisol cosecretion, including 9 with overt Cushing syndrome. An iPA was present in 85 patients (34%), and 71 patients (29%) were not evaluated for cortisol excess. Patients with cortisol cosecretion were older compared to patients with iPA (58.0 years [53.0–65.5] versus 54.0 years [47.0–59.0]; $P<0.001$) and less frequently male (49% versus 72%; $P=0.004$). All patients were hypertensive, and 44% took ≥3 antihypertensive drugs. The median duration of hypertension at diagnosis of PA was 10.2 years (2.2–18.8) for patients with cortisol cosecretion and 6.2 years (0.6–14.3) for iPA ($P=0.026$) without relevant differences in blood pressure. Metabolic comorbidities were present in 140 (56%) cases and tended to be more common in patients with cortisol cosecretion (Table 1).

Data on MACE at baseline were available for 96%; 39 patients (16%) had prior MACE, with 5 patients having 2. Median time from MACE to PA diagnosis was 31.0 months (8.0–95.5). Patients with prior MACE were older (58.0 [54.0–65.05] versus 54.0 [49.0–61.0]; $P=0.001$), more often male (79% versus 60%; $P=0.019$), had longer durations of hypertension (16.2 years [9.6–24.5] versus 8.6 [1.8–15.8], $P<0.001$), and higher rates of diabetes (36% versus 18%; $P=0.017$; Table S3) compared with those with no past MACE. The rates of MACE did not differ between patients with cortisol cosecretion and iPA ($P=0.656$). A detailed description of the type of MACE at baseline is provided in Table S4.

Biochemical and Radiological Evaluation of BMAD With PA at Baseline

Plasma aldosterone concentration was higher in patients with iPA compared with patients with cortisol cosecretion (240.0 ng/L [152.9–383.0] versus 184.0 ng/L [105.5–283.0]; $P=0.008$). Overall, hypokalemia was present in 64% of cases, with slightly though not significantly higher rates in patients with iPA versus those with cortisol cosecretion (62% versus 49%, $P=0.096$; Table 1).

Patients with cortisol cosecretion had slightly larger nodules compared to patients with iPA (24.0 mm [18.0–29.8] versus 18.0 mm [15.0–24.0]; $P<0.001$). Nodule size showed a weak positive correlation with age ($r=0.23$ [95% CI, 0.10–0.35]; $P<0.001$) and post-DST cortisol ($r=0.33$ [95% CI, 0.19–0.46]; $P<0.001$), and a weak negative correlation with ACTH ($r=-0.22$ [95% CI, −0.37 to −0.06]; $P=0.007$; Table S5). Genetic analyses were available in 24 patients (10%) with details shown in Table S6.

AVS was performed in 183 patients (73%); 48% (88/183) suggested unilateral and 42% (77/183) bilateral disease, while 10% (18/183) showed inconclusive results (13% of patients with cortisol cosecretion versus 7.4% of patients with iPA; Table 1). Patients with unilateral PA were younger (53.0 [48.0–58.0] versus 56.0 [50.0–63.0]; $P=0.020$), had higher plasma aldosterone concentration and aldosterone-to-renin ratio, and were more often hypokalemic compared with bilateral PA (79% versus 54%; $P<0.001$). No significant differences were found in hypertension severity or baseline MACE in patients with unilateral versus bilateral PA at AVS (Table S7).

Detailed AVS characteristics, including selectivity index, lateralization index, and contralateral ratio values under basal and ACTH-stimulated conditions, did not significantly differ between patients with cortisol cosecretion and those with iPA, and are summarized in Table S8.

Treatment Modality and Overall Outcome Across Treatment Groups

Data on treatment modality were available in 240 out of 249 (96%) cases. Most patients received multiple therapies (Figure 1). Overall, adrenalectomy was performed in 89 patients (89/240, 37%), including 83 unilateral, 5 subtotal bilateral, and one complete bilateral adrenalectomy. Of these, 81 patients (81/89, 91%) underwent adrenalectomy alone, while 7 (7/89, 8%) also received MRA therapy, and one (1/89, 1%) was additionally treated with steroidogenesis inhibitors. During the entire observation period, 180 patients (180/240, 75%) were treated with MRA, at least for some time. In total, 5 patients (2%) received steroidogenesis inhibitors (2 in combination with MRA therapy, 1 in combination with adrenalectomy), while 6 (2%) did not receive any specific therapy.

Follow-up data after medical or surgical intervention were available for 235 cases (235/249, 94%), with a median follow-up duration of 47.0 months (18.0–96.0) since diagnosis. In total, 2 patients died during follow-up (one due to a MACE and the other from pancreatic cancer). Of those with available data, most patients had controlled (ie, <140/90 mm Hg) blood pressure (176/220, 80%). Around a third (82/233, 35%) took at least 3 antihypertensive drugs. A new MACE occurred in 22 (10%) out of 223 patients with available information (Figure 2). Most patients had at least one metabolic comorbidity (163/231, 71%).

Table 1. Characteristics of Patient Cohort and Comparison Between Patients With Cortisol Cosecretion Versus Isolated Primary Aldosteronism

Variable	BMAD and PA	BMAD and PA with cortisol cosecretion	BMAD and isolated PA	*P* value, PA with cortisol cosecretion vs isolated PA
N, (%)	249	93 (52%)	85 (47%)	
Baseline evaluation				
Males/total, n (%)	154/249 (62%)	46/93 (49%)	61/85 (72%)	0.004*
Age, y, median (IQR)	55.0 (50.0–62.5)	58.0 (53.0–65.5)	54.0 (47.0–59.5)	<0.001*
BMI, kg/m², median (IQR)	28.2 (24.7–32.0)	28.1 (24.2–32.0)	29.3 (24.7–32.9)	0.342
Clinical presentation at baseline				
≥3 antihypertensive medications/available data, n/n (%)	110/249 (44%)	36/93 (39%)	37/85 (44%)	0.544
Antihypertension medication (defined daily dose), median (IQR)	2.8 (1.5–4.7)	2.5 (1.3–4.0)	3.0 (1.6–4.8)	0.180
Systolic blood pressure, mm Hg, median (IQR)	146 (133–159)	145 (133–159)	147 (134–160)	0.615
Diastolic blood pressure, mm Hg, median (IQR)	87 (80–96)	87 (80–95)	86 (80–96)	0.846
Duration of hypertension at referral, y, median (IQR)	9.9 (2.3–17.1)	10.2 (2.2–18.8)	6.2 (0.6–14.3)	0.026*
Overt Cushing syndrome/available data, n/n (%)	9/178 (5%)	9/93 (9.7%)	0/85 (0%)	0.003*
MACE/available data, n/n (%)	39/240 (16%)	13/87 (15%)	10/84 (12%)	0.656
Comorbidities and target organ damage at baseline				
At least one comorbidity, n/n (%)	140/249 (56%)	58/93 (62%)	41/85 (48%)	0.070
Diabetes/available data, n/n (%)	52/249 (21%)	22/93 (24%)	16/85 (19%)	0.468
Dyslipidemia/available data, n/n (%)	107/249 (43%)	45/93 (48%)	31/85 (36%)	0.130
Obesity/available data, n/n (%)	88/249 (35%)	30/92 (31%)	33/78 (42%)	0.206
Left ventricular hypertrophy/available data, n/n	58/129 (45%)	16/41 (39%)	20/48 (42%)	0.832
Chronic kidney disease/available data, n/n (%)	37/249 (15%)	13/93 (14%)	8/85 (9%)	0.365
Microalbuminuria/available data, n (%)	62/144 (43%)	21/49 (43%)	19/46 (41%)	1.000
Biochemical data at baseline				
Plasma aldosterone concentration, ng/L, median (IQR)	238.6 (133.0–367.0)	184.0 (105.5–283.0)	240.0 (152.9–383.0)	0.008*
Plasma renin concentration, mU/L, median (IQR)	2.5 (1.3–4.9)	2.5 (1.6–5.0)	2.5 (1.6–5.3)	0.934
Aldosterone-to-renin ratio, ng/L per mU/L, median (IQR)	79.0 (38.9–214.0)	54.8 (31.2–130.8)	81.3 (36.8–254.5)	0.062
Hypokalemia/available data, n/n (%)	157/247 (64%)	45/92 (49%)	52/84 (62%)	0.096
Morning serum cortisol on DST, µg/dL, median (IQR)	2.0 (1.2–3.9)	3.6 (2.7–6.6)	1.1 (0.9–1.4)	<0.001*
ACTH, pg/mL, median (IQR)	10.0 (5.0–17.0)	7.3 (4.1–11.1)	12.7 (5.0–18.4)	<0.001*
Imaging studies and AVS				
Diameter of largest adrenal nodule, mm, median (IQR)	20.0 (15.0–27.0)	24.0 (18.0–29.5)	18.0 (15.0–24.0)	<0.001*
AVS performed/available data, n (%)	183/249 (73%)	64/93 (69%)	68/85 (80%)	0.123
AVS indicating unilateral disease/available data, n/n (%)	88/183 (48%)	24/64 (38%)	38/68 (56%)	0.038*
AVS indicating bilateral disease/available data, n/n (%)	77/183 (42%)	32/64 (50%)	25/68 (37%)	0.160
AVS with inconclusive results/available data, n/n (%)	18/183 (9.8%)	8/64 (13%)	5/68 (7.4%)	0.388
Treatment modality after diagnosis of BMAD and PA				
Treatment modality described/total, n/n (%)	240/249 (96%)	91/93 (98%)	81/85 (95%)	0.427
Adrenalectomy/available data, n (%)	89/240 (37%)	28/91 (31%)	33/81 (41%)	0.203
Unilateral adrenalectomy/available data, n (%)	83/89 (93%)	27/28 (96%)	30/33 (91%)	0.618
Bilateral adrenalectomy/available data, n/n (%)	1/89 (1.1%)	0/28 (0%)	1/33 (3.0%)	1.000
Subtotal bilateral adrenalectomy/available data, n/n (%)	5/89 (5.6%)	1/28 (3.7%)	2/33 (6.1%)	1.000
Median duration until adrenalectomy, mo, median (IQR)	5.0 (2.5–17.5)	4.0 (3.0–12.5)	5.0 (2.0–23.0)	0.624
MRA therapy/available data, n/n (%)	129/240 (54%)	58/91 (64%)	49/81 (60%)	0.753
Steroidogenesis inhibitors/available data, n (%)	5/240 (2.1%)	5/91 (5.5%)	0/81 (0.0%)	0.061
No specific therapy/available data, n (%)	6/240 (2.5%)	5/91 (5.5%)	1/81 (1.2%)	0.215
Unknown treatment modality/total, n/n (%)	9/249 (3.6%)	2/93 (2.2%)	4/85 (4.7%)	0.427

ACTH indicates adrenocorticotropic hormone; AVS, adrenal venous sample; BMAD, bilateral macronodular adrenocortical disease; BMI, body mass index; DST, dexamethasone suppression test; IQR, interquartile range; MACE, major adverse cardiovascular event; MRA, mineralocorticoid receptor antagonist; and PA, primary aldosteronism.

*Significant *P* value.

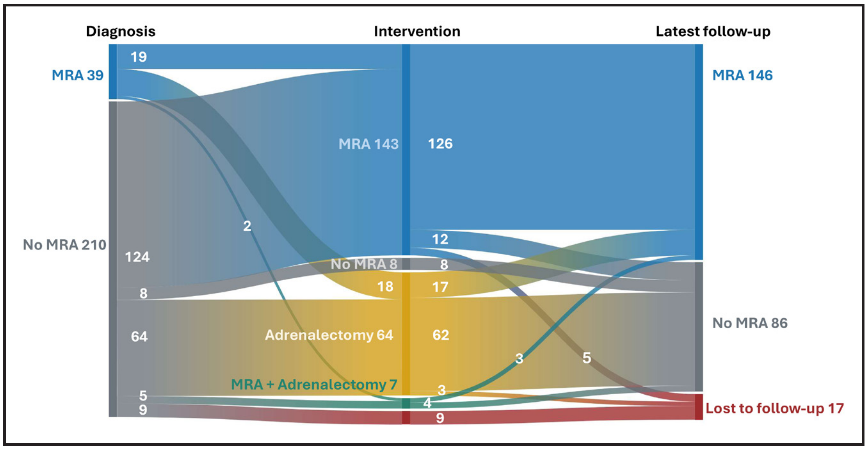

Figure 1. Overview of different treatment modalities over time. Sankey diagram indicating different treatment modalities at time of diagnosis, intervention (ie, time of adrenalectomy or initiation of mineralocorticoid receptor antagonist [MRA] therapy), and latest available follow-up. No MRA refers to any nontargeted therapy, including any antihypertensive medication other than MRA (also includes steroidogenesis inhibitors).

Clinical and Biochemical Outcome in Surgically Versus MRA-Treated Patients

To assess treatment impact, we compared patients who only underwent adrenalectomy (ie, who never received MRA therapy [also not before PA diagnosis], n=50) with patients who only received MRA therapy (initiated at baseline or the intervention timepoint and never underwent adrenalectomy, n=124). Importantly, other antihypertensive medications were used in both groups as needed. Patients treated with steroidogenesis inhibitors or any other nonspecific therapy were excluded to avoid bias (Table 2; Figure 3).

Surgically treated patients were younger than MRA-treated patients (52.0 years [47.8–56.0] versus 57.5 years [52.0–63.0]; P<0.001), but groups were similar in sex, body mass index, cortisol cosecretion, plasma aldosterone concentration, plasma renin concentration, aldosterone-to-renin ratio, and baseline metabolic comorbidities and MACE. AVS was performed in 92% surgically versus 67% MRA-treated patients (P<0.001). Of these, 89% were suggestive of unilateral disease in surgically treated patients versus 19% of MRA-treated patients (P<0.001; Table 2).

AVS data were available for most surgically treated patients and showed substantial overlap across outcome groups. When stratified by biochemical outcome after adrenalectomy, no significant differences were observed in selectivity index or lateralization index either under basal conditions or after ACTH stimulation, although contralateral suppression tended to be more frequent in patients achieving complete biochemical success (Table S9).

When stratified by clinical outcome, the contralateral ratio differed significantly across groups under basal conditions, with higher values observed in patients with absent clinical success (P=0.005; Table S10). In contrast, selectivity index and lateralization index did not differ significantly across clinical outcome categories. Overall, AVS parameters showed limited ability to predict clinical or biochemical outcome after adrenalectomy in this cohort.

During a median follow-up of 36 months (9.0–81.5) after MRA initiation and 18 months (5.8–70.3; P=0.207)

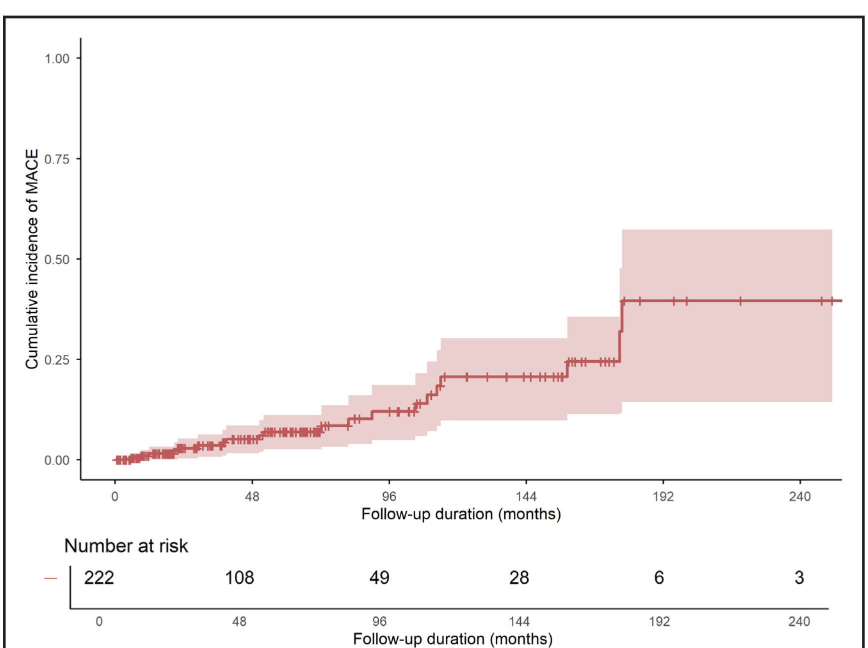

Figure 2. Cumulative incidence of major adverse cardiovascular events in the entire study cohort. Unadjusted cumulative incidence of major cardiovascular events (MACE) since diagnosis of primary aldosteronism (PA). Overall, cardiovascular status at the latest follow-up was available for 222 patients. However, the date of MACE occurrence was unknown in one case; thus, the graph shows 222 patients. The shaded areas represent the 95% CIs. The number of patients at risk is displayed at each time point.

Table 2. Comparison Between Patients Who Only Underwent Adrenalectomy and Patients Who Only Received MRA Therapy at Baseline and Follow-Up

Variable	Surgically treated patients	MRA-treated patients	*P* value
N	50	124	…
Demographic information			
Males/total, n (%)	33/50 (66%)	78/124 (63%)	0.731
Age, y, median (IQR)	52.0 (47.8–56.0)	57.5 (52.0–63.0)	<0.001*
BMI, kg/m², median (IQR)	26.8 (23.9–30.6)	27.6 (24.7–30.8)	0.298
Baseline evaluation			
≥3 antihypertensive medications/available data, n/n (%)	18/50 (36%)	52/124 (42%)	0.499
Antihypertension medication (defined daily dose), median (IQR)	2.5 (1.3–4.9)	2.7 (1.3–4.5)	0.895
Systolic blood pressure, mm Hg, median (IQR)	143 (130–160)	147 (134–159)	0.363
Diastolic blood pressure, mm Hg, median (IQR)	87 (80–92)	86 (80–99)	0.627
Duration of hypertension at referral, years, median (IQR)	6.0 (1.6–10.5)	10.0 (1.9–19.4)	0.046*
Cortisol cosecretion/available data, n/n (%)	16/36 (44%)	47/89 (53%)	0.423
Overt Cushing syndrome/available data, n/n (%)	5/36 (14%)	0/89 (0%)	0.002*
MACE/available data, n/n (%)	6/50 (12%)	17/118 (14%)	0.808
Comorbidities and target organ damage at baseline			
At least one comorbidity/available data, n/n (%)	27/50 (54%)	67/124 (54%)	1.000
Diabetes/available data, n/n (%)	9/50 (18%)	22/124 (18%)	1.000
Dyslipidemia/available data, n/n (%)	19/50 (38%)	55/124 (44%)	0.500
Left ventricular hypertrophy/available data, n/n	10/28 (36%)	33/70 (47%)	0.370
Chronic kidney disease/available data, n/n (%)	9/50 (18%)	17/124 (14%)	0.487
Microalbuminuria/available data, n (%)	14/27 (52%)	27/74 (36%)	0.178
Biochemical data at baseline			
Plasma aldosterone concentration, ng/L, median (IQR)	247.2 (101.3–518.3)	189.0 (119.0–314.0)	0.140
Plasma renin concentration, mU/L, median (IQR)	2.3 (1.0–5.6)	2.5 (1.5–4.5)	0.786
Aldosterone-to-renin ratio, ng/L per mU/L, median (IQR)	84.2 (37.3–266.1)	64.0 (34.5–158.5)	0.472
Hypokalemia/available data, n/n (%)	38/50 (76%)	76/123 (62%)	0.080
DST in µg/dL, median (IQR)	1.8 (1.2–3.8)	1.9 (1.1–3.5)	0.563
ACTH in pg/mL, median (IQR)	10.0 (3.0–13.2)	11.0 (6.0–17.9)	0.177
Imaging studies and AVS			
Diameter of largest adrenal nodule, mm, median (IQR)	19.5 (16.0–30.0)	19.0 (15.0–24.0)	0.117
AVS performed/available data, n (%)	46/50 (92%)	83/124 (67%)	<0.001*
AVS indicating unilateral disease/available data, n/n (%)	41/46 (89%)	16/83 (19%)	<0.001*
AVS indicating bilateral disease/available data, n/n (%)	4/46 (8.7%)	53/83 (64%)	<0.001*
AVS with inconclusive results/available data, n/n (%)	1/46 (2.2%)	14/83 (17%)	0.019*
Biochemical data at last follow-up			
Plasma aldosterone concentration, ng/L, median (IQR)	157.6 (73.4–303.0)	168.0 (74.8–311.5)	<0.001*
Plasma renin concentration, mU/L, median (IQR)	9.9 (3.3–23.3)	10.0 (3.7–23.3)	0.188
Aldosterone-to-renin ratio, ng/L per mU/L, median (IQR)	15.8 (4.5–41.1)	16.1 (4.6–40.3)	0.008*
Plasma renin concentration in mU/Lnot suppressed/available data, n/n (%)†	20/45 (44%)	44/82 (54%)	0.357
Hypokalemia/available data, n/n (%)	3/49 (6%)	8/113 (7%)	1.000
Comorbidities and target organ damage at last follow-up			
Follow-up duration since intervention, mo, median (IQR)	18.0 (5.8–70.3)	36.0 (9.0–81.5)	0.207
Patients with blood pressure >140/90 mm Hg/available data, n/n (%)	10/48 (21%)	26/118 (22%)	1.000
≥3 antihypertensive medications/available data, n/n (%)	7/50 (14%)	59/124 (48%)	<0.001*
Antihypertension medication (defined daily dose), median (IQR)	2.0 (0.0–3.0)	3.0 (1.7–4.2)	<0.001*
New MACE during entire follow-up/available data, n/n (%)	3/47 (6.4%)	10/120 (8%)	1.000
New MACE during the first 18 mo of follow-up/available data, n/n (%)	1/47 (2.1%)	2/120 (1.7%)	1.000

(*Continued*)

Table 2. Continued

Variable	Surgically treated patients	MRA-treated patients	*P* value
Diabetes/available data, n/n (%)	15/49 (31%)	37/124 (30%)	1.000
Dyslipidemia/available data, n/n (%)	26/49 (53%)	79/124 (64%)	0.228
Left ventricular hypertrophy/available data, n/n (%)	7/16 (44%)	33/57 (58%)	0.398
Chronic kidney disease/available data, n/n (%)	16/49 (33%)	24/124 (19%)	0.073
New microalbuminuria/available data, n/n (%)	9/22 (41%)	17/58 (29%)	0.423
Hypokalemia/available data, n/n (%)	3/49 (6.1%)	7/120 (5.8%)	1.000

Patients who underwent adrenalectomy and received MRA therapy were excluded. Likewise, patients who received steroidogenesis inhibitors or any nonspecific therapy were also excluded from the analysis. ACTH indicates adrenocorticotropic hormone; AVS, adrenal venous sample; BMI, body mass index; DST, 1 mg dexamethasone suppression test; IQR, interquartile range; MACE, major adverse cardiovascular events; MRA, mineralocorticoid receptor antagonists; and PAMO, primary aldosteronism medical treatment outcome.

*Significant *P* value.

†Suppressed plasma renin concentration is defined as <10 mU/L, in accordance with the PAMO study.[48]

following adrenalectomy, MACE occurred in 8% and 6% of patients, respectively (P=1.000; Table 2; Figure 3). Likewise, rates of uncontrolled blood pressure and target organ damage were similar across groups. However, more MRA-treated patients (48%) required ≥3 antihypertensives compared to surgically treated patients (14%, P<0.001; Table 2), who also showed a significantly lower antihypertensive drug burden at follow-up, as reflected by a reduced DDD (2.0 versus 3.0; P<0.001; Table 2).

A detailed description of the type of MACE at follow-up is provided in Table S11. In a multivariable Cox regression analysis, treatment modality (ie, surgery only versus MRA therapy only) was not associated with a substantial difference regarding MACE occurrence, neither alone (HR for surgery, 1.7 [95% CI, 0.4–7.8]; P=0.481), nor after adjusting for age and sex (Table S12).

According to the PASO criteria, clinical success was considered complete in 26% (13/50), partial in 62% (31/50), and absent in 12% (6/50) of surgically treated patients. Biochemical success was considered complete in 71% (34/48 with available biochemical follow-up data), partial in 25% (12/48), and absent in 4% (2/48) of patients. As expected, plasma aldosterone concentration and aldosterone-to-renin ratio significantly decreased

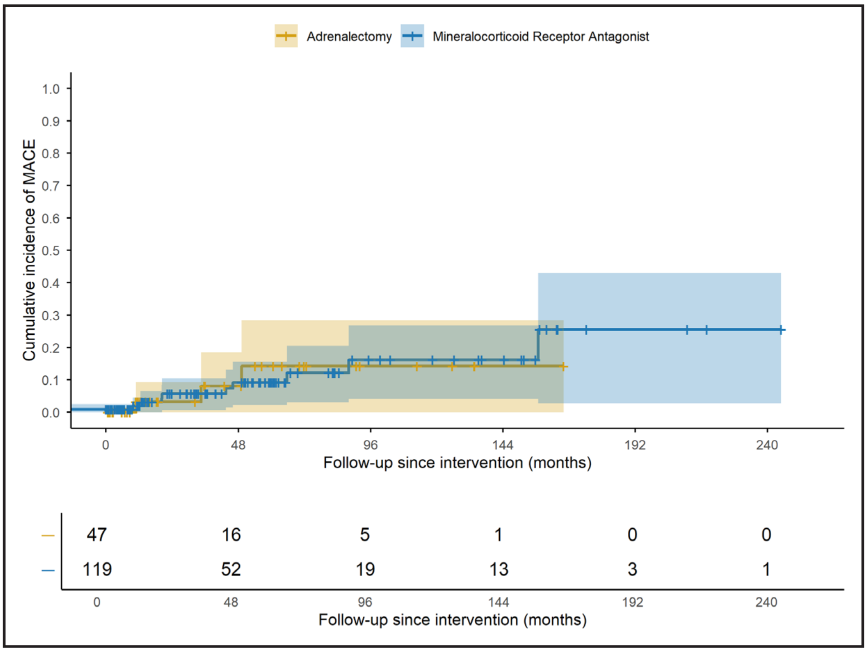

Figure 3. Cumulative incidence of major adverse cardiovascular events following targeted intervention.
Cumulative unadjusted incidence of major cardiovascular events (MACE) since initiation of therapy. The graph shows patients who were treated exclusively with surgery (ie, never received mineralocorticoid receptor antagonist [MRA] therapy) in yellow, and those who received only MRA therapy (ie, never underwent surgical intervention) in blue. Patients who underwent both surgical and MRA therapy, as well as those who received steroidogenesis inhibitors or no targeted therapy, were excluded from the analysis. Of the 124 patients in the MRA-only group, 4 patients were excluded due to unknown MACE status, and one was excluded because they experienced a MACE after diagnosis but before the initiation of MRA therapy. Of the 50 patients in the adrenalectomy–only group, 3 patients with unknown MACE status were excluded from the analysis. Adrenalectomy was unilateral in 45 patients, while 2 received a subtotal bilateral adrenalectomy. The shaded areas represent the 95% CIs. The number of patients at risk is displayed at each time point. The log-rank test showed no significant difference in the cumulative incidence of major cardiovascular events between the adrenalectomy and MRA groups (χ^2=0.0, df=1, P=0.997).

while plasma renin concentration increased following adrenalectomy (Table S13). At the last follow-up, most patients (94%) no longer had hypokalemia, without a significant difference between the 2 groups (P=1.000; Table 2).

We did not observe significant differences in plasma renin concentration or aldosterone-to-renin ratio levels at the last follow-up between patients treated with surgery and those on MRA therapy. As expected, plasma aldosterone concentrations were significantly lower in the surgical group compared to the MRA group (157.6 versus 168.0; P<0.001). Notably, 54% (44/82) of patients in the MRA group had a nonsuppressed plasma renin concentration (ie, ≥10 mU/L), compared with 44% (20/45) in the surgical group (P=0.357; Table 1).

When analyzed regardless of treatment approach, no substantial difference in the occurrence of MACE was detected between patients with cortisol co-secretion versus those with isolated PA (Figure 4).

DISCUSSION

This is the first multicenter and largest cohort study on patients with BMAD and PA. This population had a high proportion of associated cardiovascular and metabolic comorbidities already at baseline. Notably, hypertension was present for a median duration of almost 10 years at the diagnosis of PA. This finding aligns with previous studies reporting similar diagnostic delays in the diagnosis of PA (ranging from 8 to 15 years),[25,38–40] further emphasizing the need for increased clinical awareness.

The use of multiple treatment modalities over time reflects the challenges in managing BMAD with PA and the lack of clear recommendations. Similar therapeutic dilemmas have been described in patients with bilateral adrenal tumors and cortisol excess, although surgical treatment was associated with better biochemical control and improvement in some comorbidities.[41] Our study is the first to compare treatment strategies for patients with BMAD and PA. The primary end point was MACE incidence. At baseline, 16% of patients had already experienced at least one MACE, which is consistent with prior reports from Japan, Italy, and Germany, ranging from 9.4% to 16.3%.[39,40,42] As expected, age, male sex, diabetes, and duration of hypertension were associated with MACE at baseline. Interestingly, cortisol cosecretion was not identified as an independent risk factor for MACE in this cohort. Since cortisol can cause organ damage partly via MR (mineralocorticoid receptor) activation, MRAs might offset this effect by blocking MR signaling, potentially explaining the lack of association.[43] In contrast, Nakajima et al[44] reported higher rates of cardiovascular events in patients with PA and autonomous cortisol secretion with a 1 mg-DST result of ≥3 µg/dL.[45] This discrepancy may reflect the milder cortisol cosecretion in our cohort (median, 1 mg-DST: 3.6 µg/dL). Due to the limited sample size, subanalyses by cortisol excess severity were not feasible. We observed a large proportion (around 64%) of patients with spontaneous hypokalemia, which is higher than typically reported in the literature (around 28%).[16,18] This could be attributed to more severe forms of PA being screened at our tertiary-care center.

During a relatively short follow-up duration of 3.9 years since diagnosis, 10% of all patients experienced at least one further MACE, with similar proportions following MRA (8%) or surgical (6%) treatment. In comparison, Catena et al[42] reported a MACE incidence of 18% over 7.4 years in 54 patients with PA but also found no relevant difference between treatments. In both studies, age and hypertension duration were key predictors for cardiovascular events. Our study additionally revealed similar comorbidity rates, with 80% achieving blood pressure control across

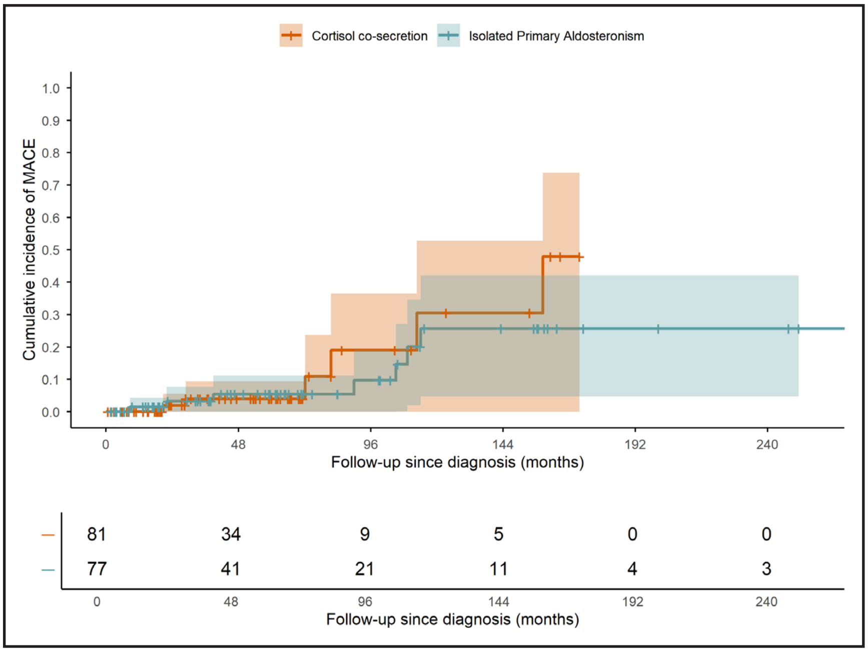

Figure 4. Cumulative incidence of major adverse cardiovascular events (MACE) in patients with and without cortisol excess.

Cumulative unadjusted incidence of MACE since diagnosis of primary aldosteronism (PA). The graph shows all 77 patients with isolated primary aldosteronism and known MACE status at follow-up in green, and all 81 patients with cortisol cosecretion and known MACE status in orange. The shaded areas represent the 95% CIs. The number of patients at risk is displayed at each time point. The log-rank test showed no significant difference in the cumulative incidence of MACE between patients with cortisol cosecretion and isolated primary aldosteronism (χ^2=0.4, df=1, P=0.536). Hazard ratio for cortisol cosecretion: 1.42 (95% CI, 0.47–4.27).

treatment groups, though surgically treated patients required fewer antihypertensive medications, potentially aiding compliance. However, the short follow-up periods and the heterogeneous patient groups (mostly bilateral PA in the MRA group and unilateral PA in the surgical group) impede drawing robust conclusions. Patients who underwent surgery were younger (52.0 versus 57.5 years), had higher rates of overt Cushing syndrome (14% versus 0%), and underwent AVS more frequently (92% versus 67%) compared with patients who received MRA therapy. The latter observation likely reflects imaging differences (eg, asymmetrical masses) and AVS availability. Unsurprisingly and in line with PA guidelines recommendations,[31] AVS more often indicated unilateral disease in surgically treated patients and bilateral disease in patients treated with MRA. Notably, most patients with AVS-confirmed unilateral disease who subsequently underwent surgery for the aldosterone-secreting nodule had complete biochemical success, supporting the use of AVS in PA with bilateral nodules. However, the complete biochemical success rate in our cohort was 71%, lower than that reported in the original PASO study.[37] This prompts caution when considering surgery in BMAD patients, even in the presence of lateralized aldosterone production on AVS. Importantly, the comparison of MACE outcomes after surgery versus MRA therapy does not represent a head-to-head comparison of equivalent interventions, but rather reflects real-world treatment strategies applied to BMAD with different PA subtypes and varying disease severity.

Furthermore, AVS is more challenging in the presence of hypercortisolism, with inconclusive results more common in those with cortisol cosecretion (13% versus 7.2%). In such cases, metanephrine, as well as other steroid biomarkers such as 18-oxocortisol, corticosterone, and 11-deoxycortisol, rather than cortisol concentrations, could be used to correct aldosterone dilution and accurately calculate the lateralization index.[33,43,44]

Our study does not support a one-size-fits-all approach to treatment. Instead, it underscores the importance of matching the right therapy to the right patient. AVS data highlight the heterogeneity of BMAD with PA, including patients with bilateral secretion, unilateral dominance, or possible steroid cosecretion. In our cohort, AVS indicated bilateral disease in 42% and unilateral in 48%. Precise subtyping is crucial to warrant optimal treatment and outcome. In BMAD, more than in unilateral PA, surgical decisions should be based on multiple factors rather than AVS alone. Moreover, the long-term reliability of AVS as a standalone tool is limited, as initially nonfunctioning nodules may later become aldosterone-producing. This progression may follow a continuum from subclinical to overt hyperaldosteronism and contribute to organ damage. Therefore, long-term follow-up is essential for all patients, including those who undergo surgery, as growing evidence suggests that nodules initially classified as nonsecreting may eventually become hormonally active over time.[46]

Strengths and Limitations

Strengths of our study include its multicenter design, large cohort, and nearly complete data set. Strict diagnostic criteria minimized the risk of misclassification, and clearly defined variables with outcome assessment based on PASO criteria are further strengths, improving generalizability. The subanalyses by cortisol cosecretion and treatment type allowed further in-depth analyses.

Limitations include the retrospective design and relatively short follow-up (median 47 months), limiting conclusions on long-term outcomes. However, the fact that 10% experienced at least one (further) MACE during this period illustrates their high risk. The overall limited number of patients and events impeded more extensive Cox regression analyses and further subanalyses to compare MACE incidence between treated and untreated patients (untreated patients n=8). Another limitation is the lack of a uniform definition of BMAD, which has evolved from being equated with Cushing syndrome and nodular adrenocortical hyperplasia[47] to defining BMAD as bilateral adrenal macronodules with possible autonomous cortisol secretion,[2] and now a primarily radiological diagnosis independent of hormonal activity. This evolving understanding highlights the need for standardized diagnostic criteria to improve comparability across studies and clinical settings. We tried to address this aspect by only including cases with precise imaging data. The lack of histopathologic diagnoses in surgically treated patients is another important limitation. Moreover, incomplete follow-up data in the MRA-treated group prevented a systematic assessment of clinical and biochemical outcomes according to the PAMO criteria.[48] As a result, direct comparison of treatment efficacy between surgical and medical management is limited. Moreover, although we report renin concentrations under MRA treatment and highlight that over half of patients had nonsuppressed plasma renin concentration (indicating potential suboptimal dosing), we were unable to fully evaluate the degree of MRA optimization in all cases. These limitations underscore the need for standardized biochemical monitoring and long-term follow-up in patients with bilateral PA receiving medical therapy.

Perspectives

Current guidelines recommend PA screening in patients with arterial hypertension and high-risk features or first-degree relatives with PA.[31] However, these guidelines are insufficiently recognized outside the endocrinology community and are inconsistently applied in clinical practice.[49] The PASO study found that shorter hypertension duration improved the chances of complete clinical success,[37] and in our study, a longer delay in diagnosis was linked to more cardiovascular events. Notably, 56% of patients already had organ damage and metabolic comorbidities at diagnosis, indicating late detection and possibly irreversible harm.

Therefore, we support broader PA screening in all patients with hypertension, as proposed by others,[18,50] to enable earlier diagnosis and timely targeted therapy. Further research, including randomized controlled trials, is warranted to evaluate the clinical and economic impact of a universal screening strategy for PA in hypertensive populations.

In conclusion, our well-characterized cohort showed that BMAD with PA is an underestimated disease entity associated with a substantial burden of cardiometabolic comorbidities. Hypertension duration and diabetes emerged as modifiable risk factors for MACE and should be managed appropriately. Our study contributes to the growing evidence base advocating for a paradigm shift in the management of hypertension, emphasizing the importance of considering and actively screening for secondary causes, particularly PA.

ARTICLE INFORMATION

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